

CSharPN

A Code-First Framework for Coloured Petri Net Modelling
with C# as Inscription Language

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PNSE'26 · Petri Nets and Software Engineering · June 2026

The gap

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What if a CPN model were just a C# file in your repo?

CPNs in 30 seconds

Two extensions used in this talk:

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- **Timed CPNs**: tokens carry timestamps, global clock advances
- **Hierarchical CPNs**: substitution transitions + port places (In, Out, I/O)

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Open source (MIT) — github.com/simontjell/CSharPN.public

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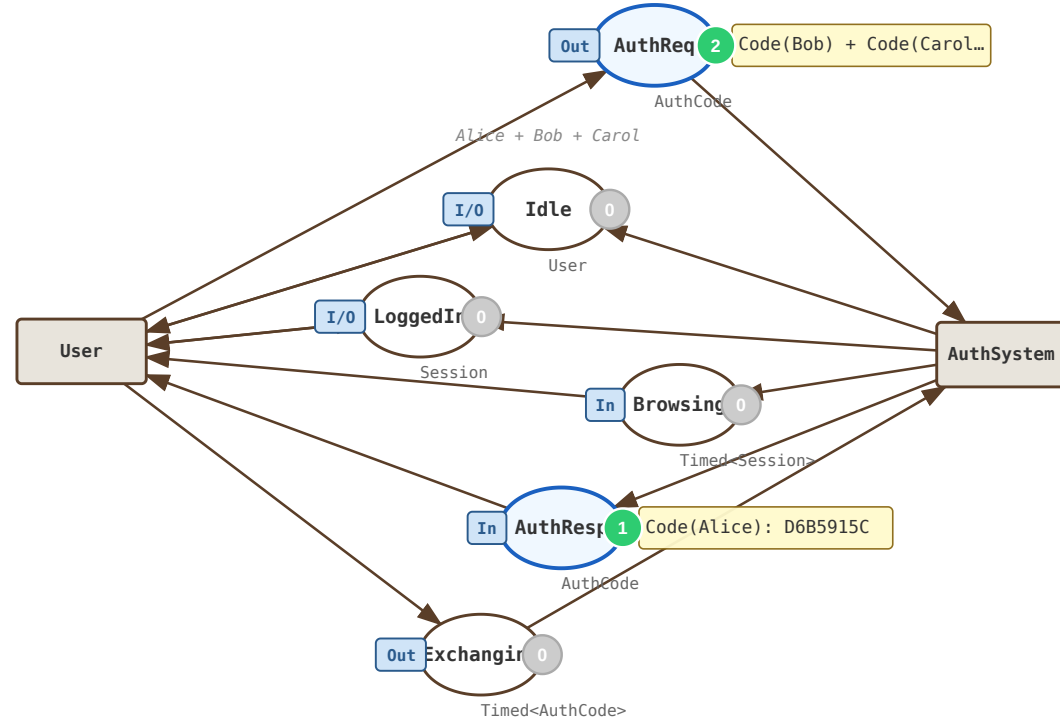
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- Interactive **Blazor visualiser** embedded in VS Code

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Running example: OAuth2 authoriz

Exercises all three features:

Top page: shared places link the two substitution transitions.

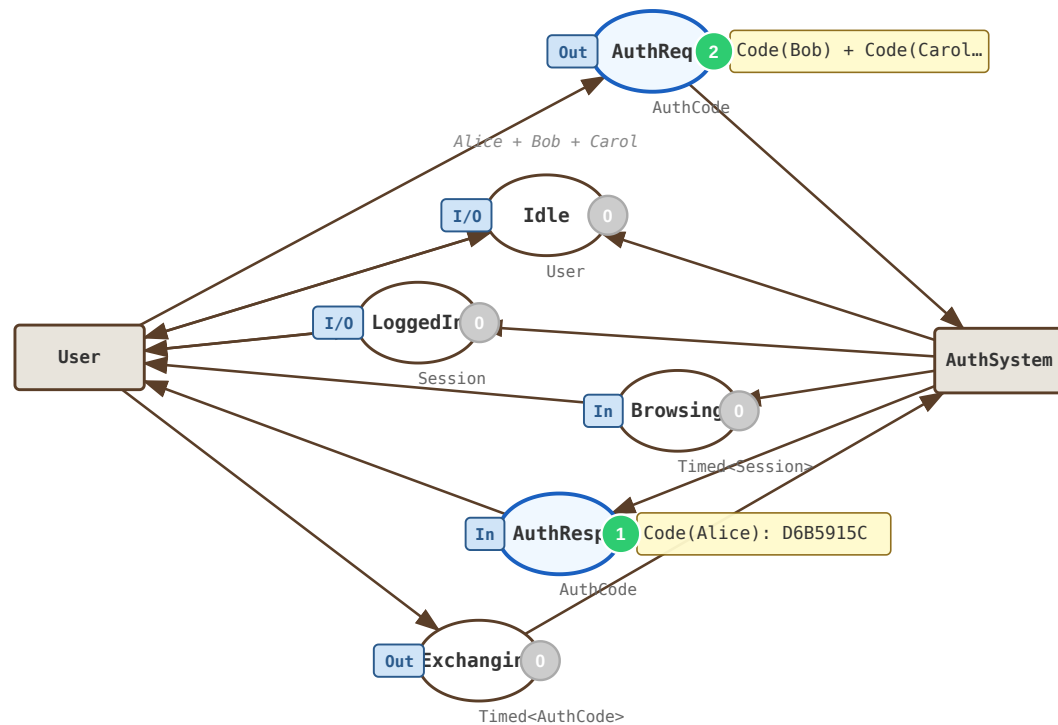


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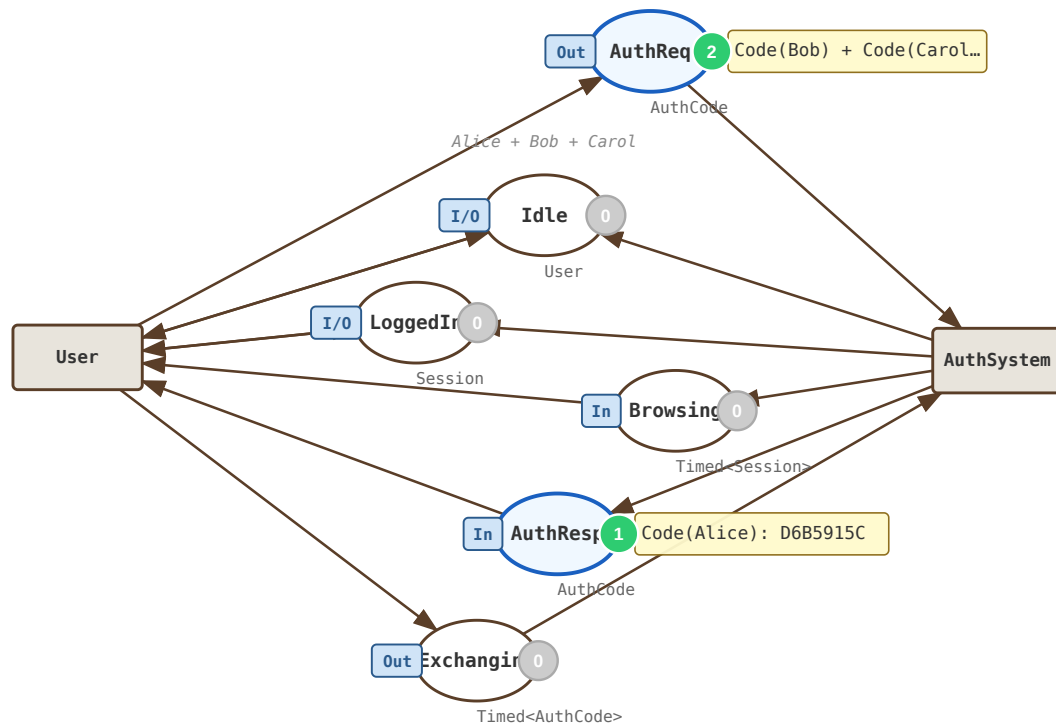


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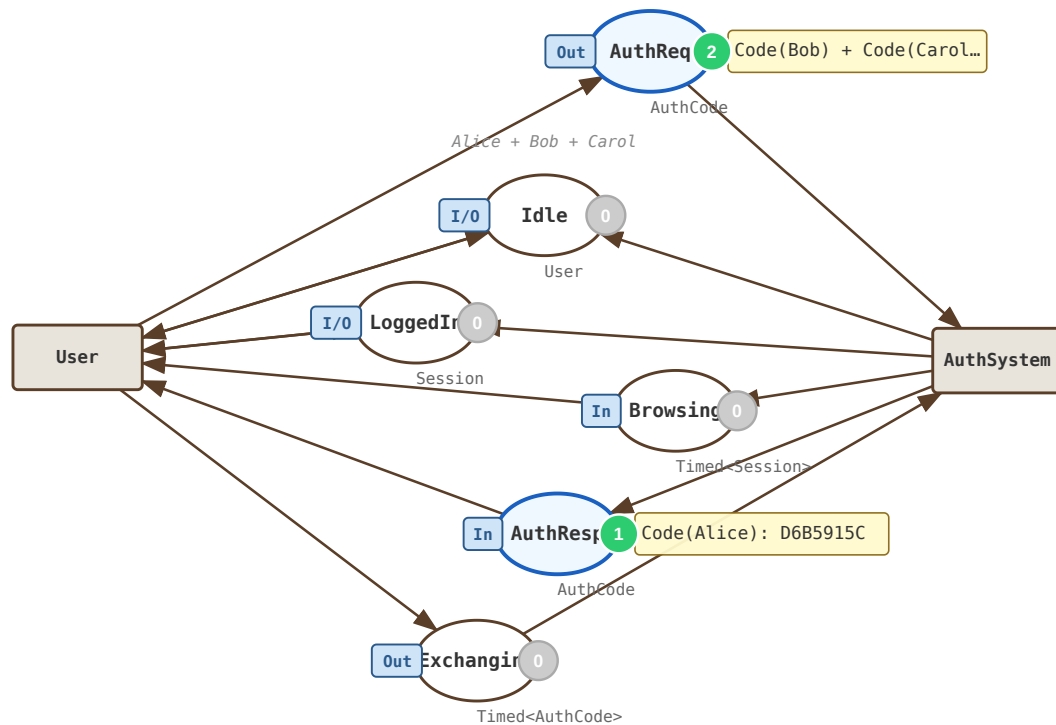


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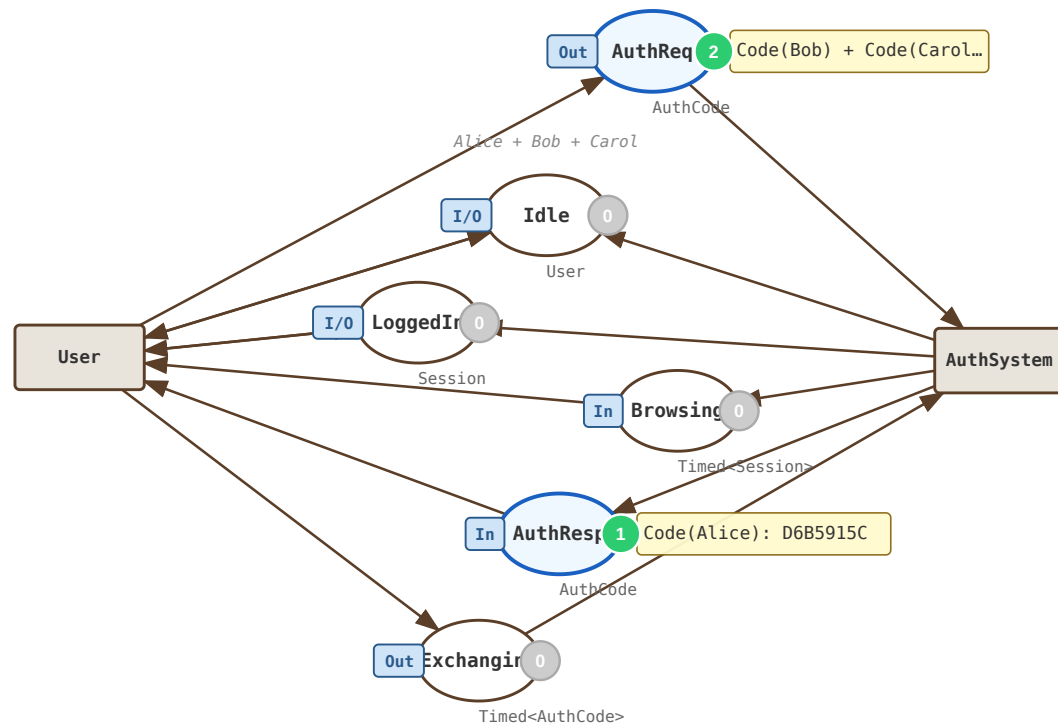


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- Two sub-pages: User & AuthSystem

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Colour sets are just C# types

Any `notnull, IEquatable<T>` type works — primitives, enums, records, classes.

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public record User(int Id, string Name) {
    public string CsrfsState => $"s{Id}";
    public int    BrowseTime => Id * 5;
}

public record AuthCode(User User, string Code, string State) {
    public bool ValidCsrfs(User u) => State == u.CsrfsState;
    public static AuthCode Create(User u, int seq) {
        var hash = SHA256.HashData(Encoding.UTF8.GetBytes($"{u.Id}:{seq}"));
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public record Session(User User, int TokenId);
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Places & multisets

```
// inside the model constructor
var authReq = AddPlace<AuthCode>("AuthReq");           // empty initial marking

var idle    = AddPlace("Idle", Multiset.Of(
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var exchg   = AddTimedPlace<AuthCode>("Exchanging");    // timed place
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var u = new Var<User>("u");

AddTransition("Login")
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    .Build();

AddTransition("ReceiveCode")
    .Input(authResp, c).Input(waiting, u)
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    .TimedInput(exchg, c).Input(nextTokId, tid)
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    .Output(loggedIn, () => new Session(c.Val.User, tid.Val))
    .TimedOutput(expiry, () => new Session(c.Val.User, tid.Val), Config.SessionTTL)
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AddTransition("SessionExpire")
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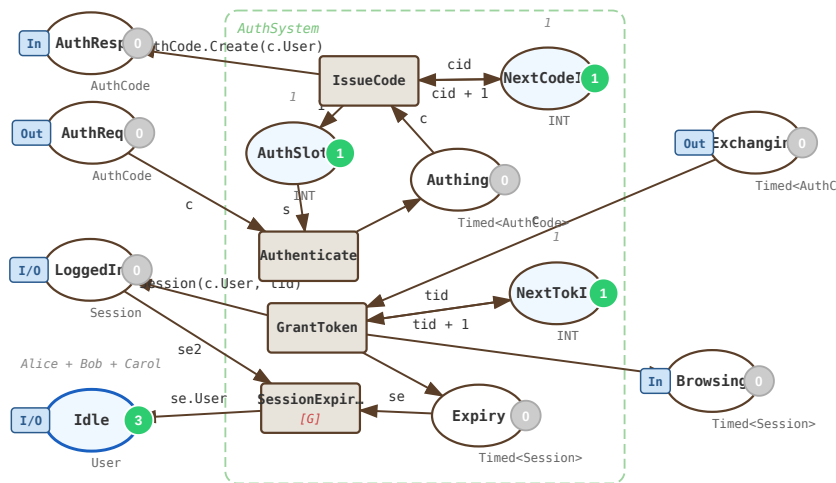
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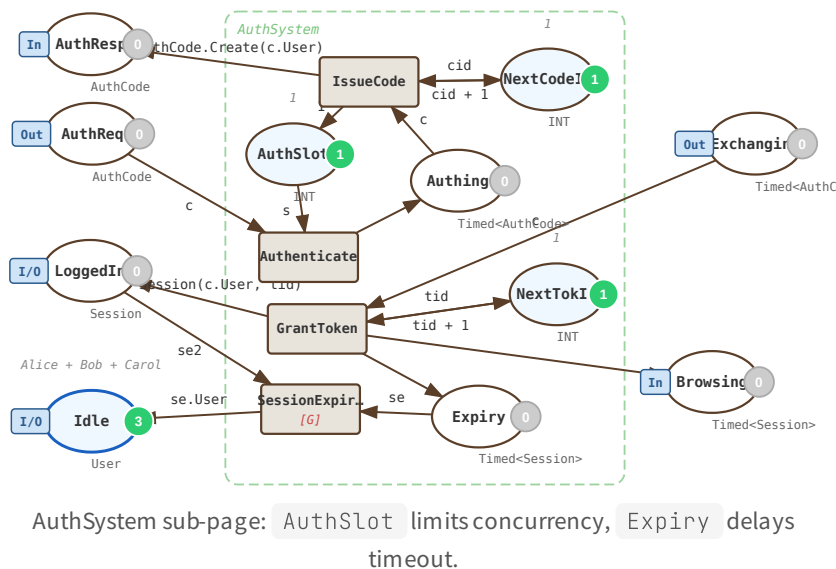
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class UserPage : CpnPage {
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// assembly
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AuthSystem sub-page: **AuthSlot** limits concurrency, **Expiry** delays timeout.

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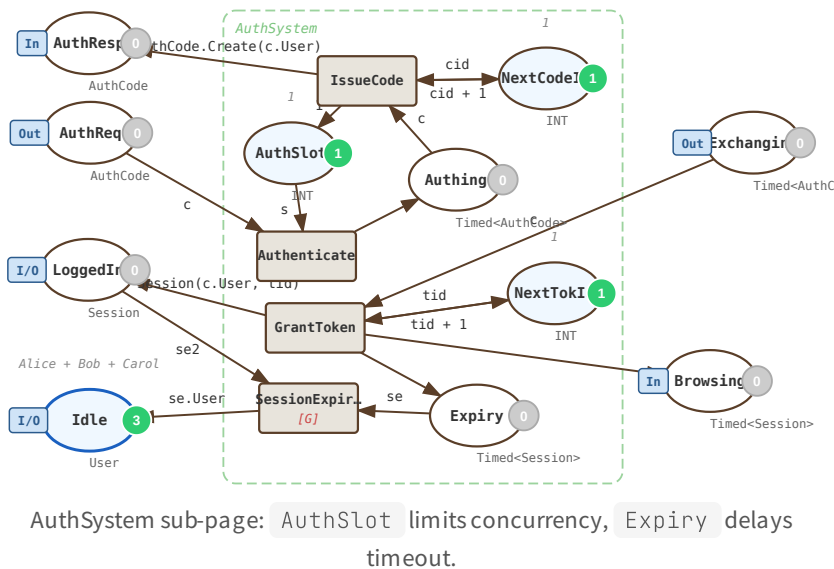
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- A sub-page is a `CpnPage` subclass

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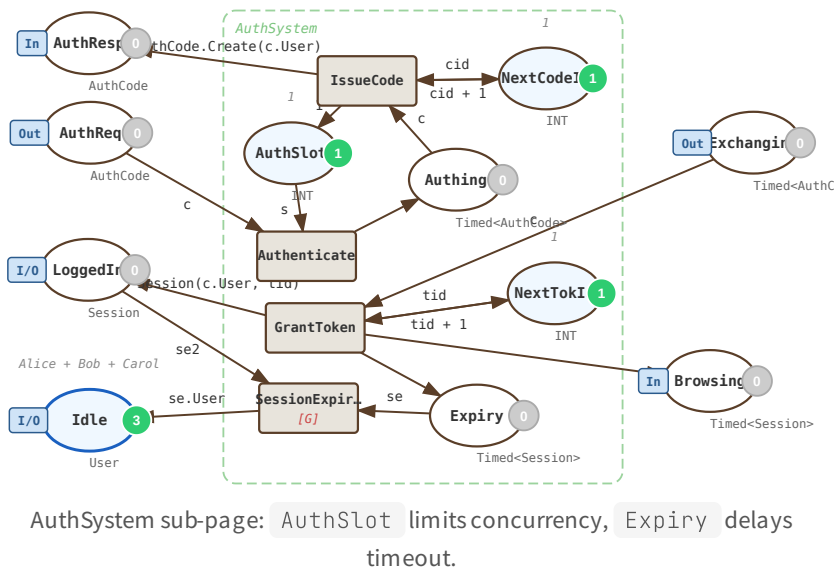
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- A sub-page is a CpnPage **subclass**
- Ports use In() / Out() / InOut() — CPN Tools style

Hierarchical CPNs

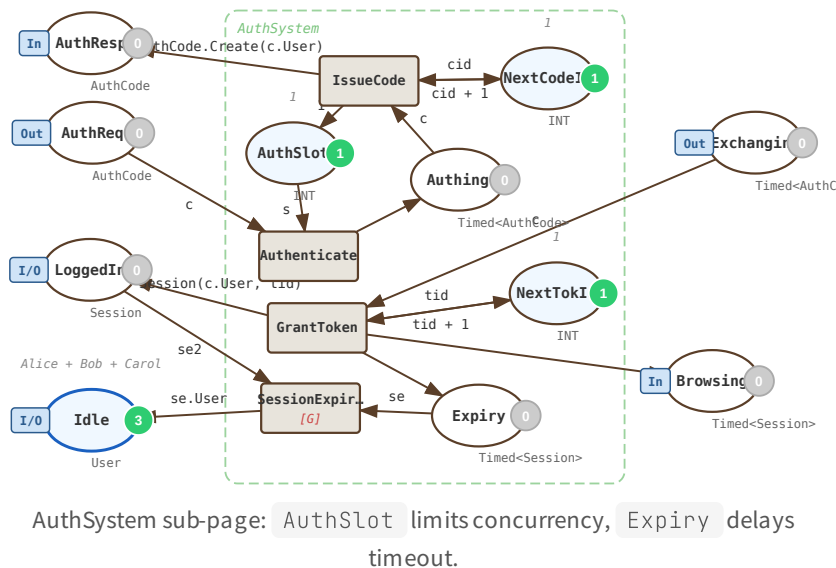
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class UserPage : CpnPage {
    public UserPage(Place<User> idle,
        Place<AuthCode> authReq, ...)
        : base("User") {
        InOut(idle);
        Out(authReq); In(authResp);
        Out(exchg);
        InOut(loggedIn); In(browsing);
        var waiting = AddPlace<User>("Waiting");
        // Login, ReceiveCode, Logout
    }
}
// assembly
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- AddSubPage wires it into the parent

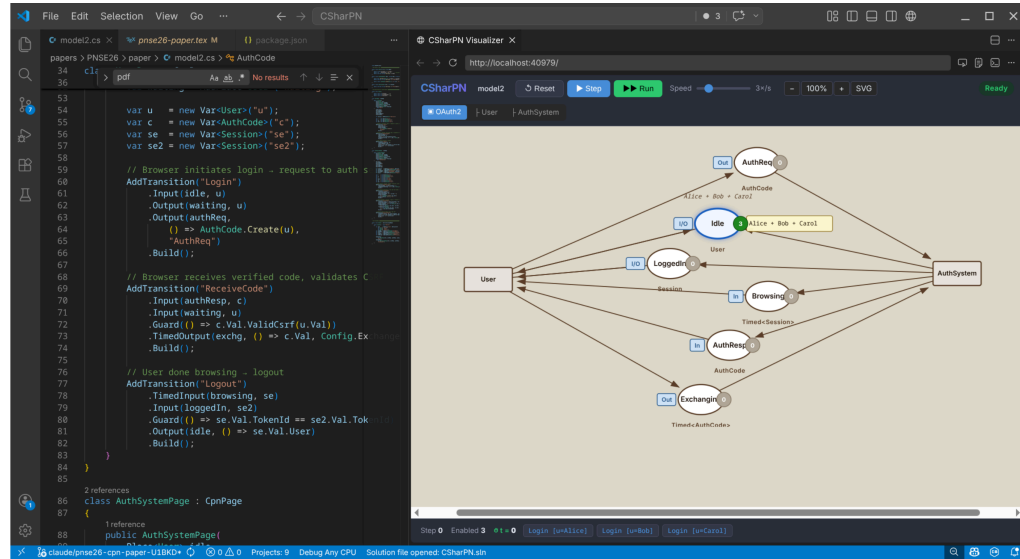
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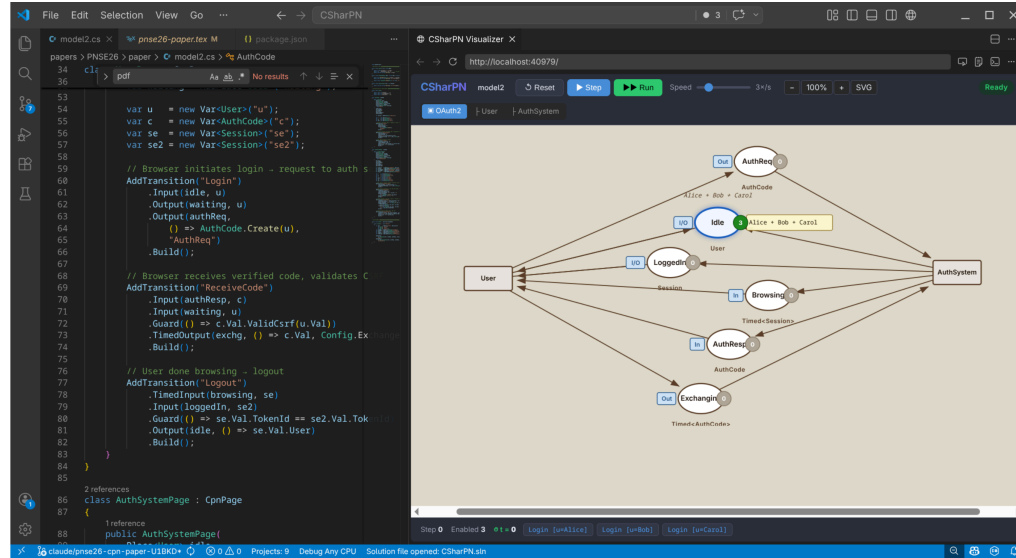
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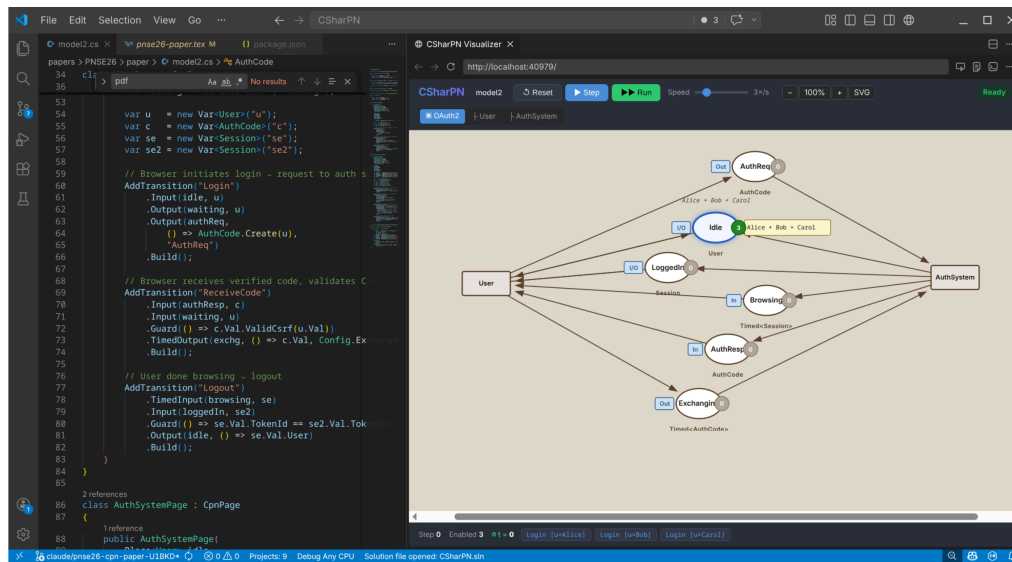
Visualiser & VS Code integration

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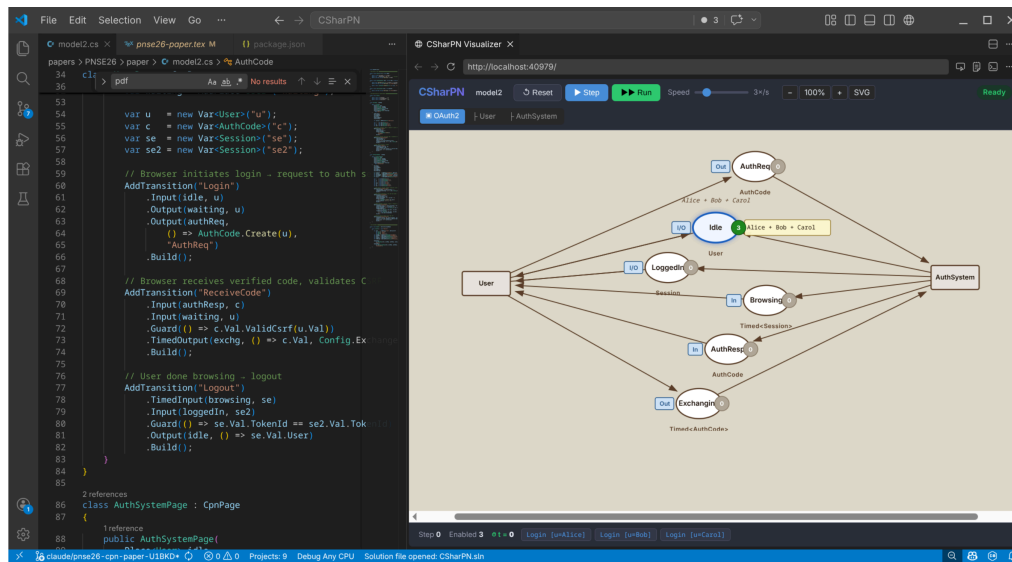
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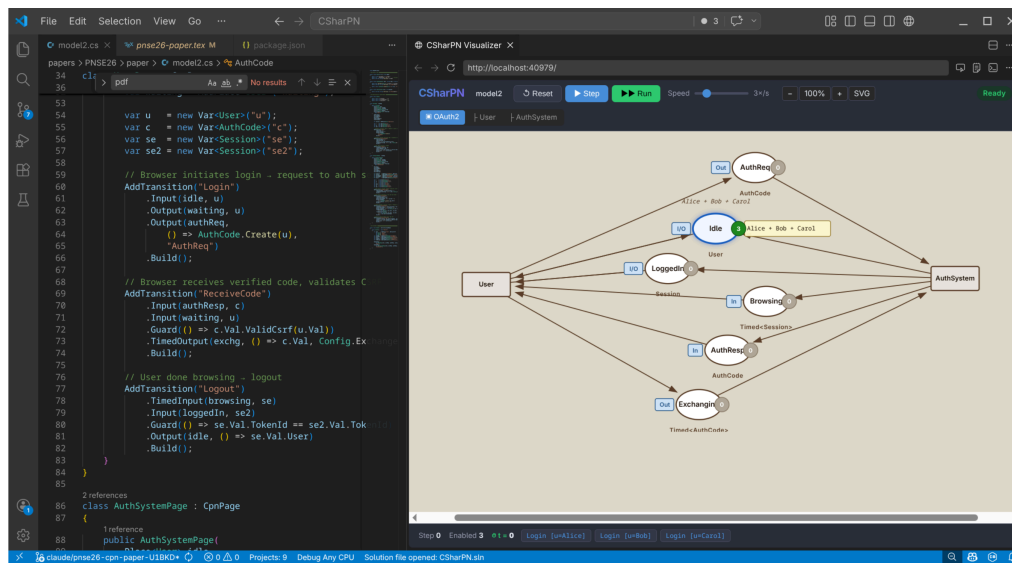
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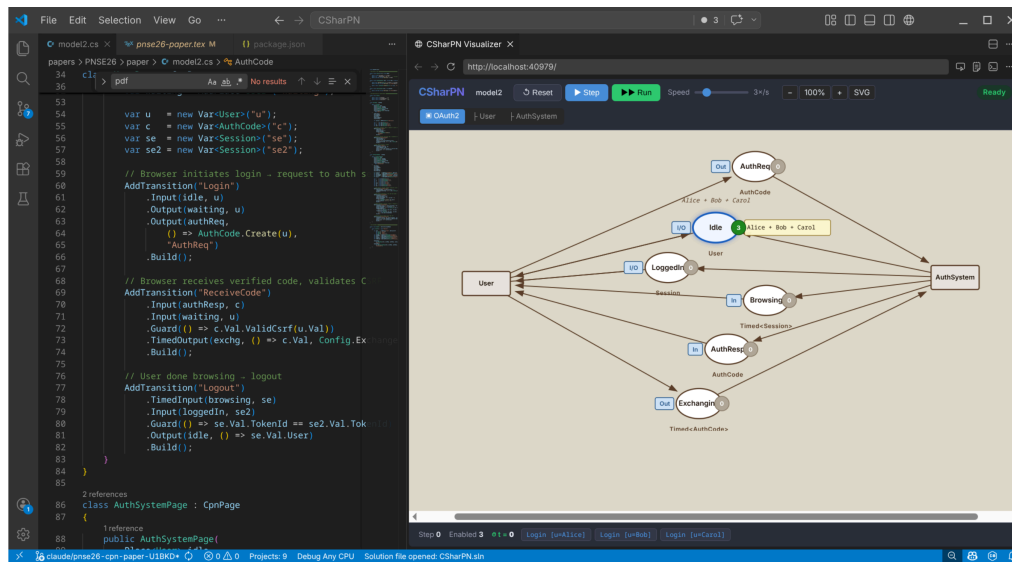
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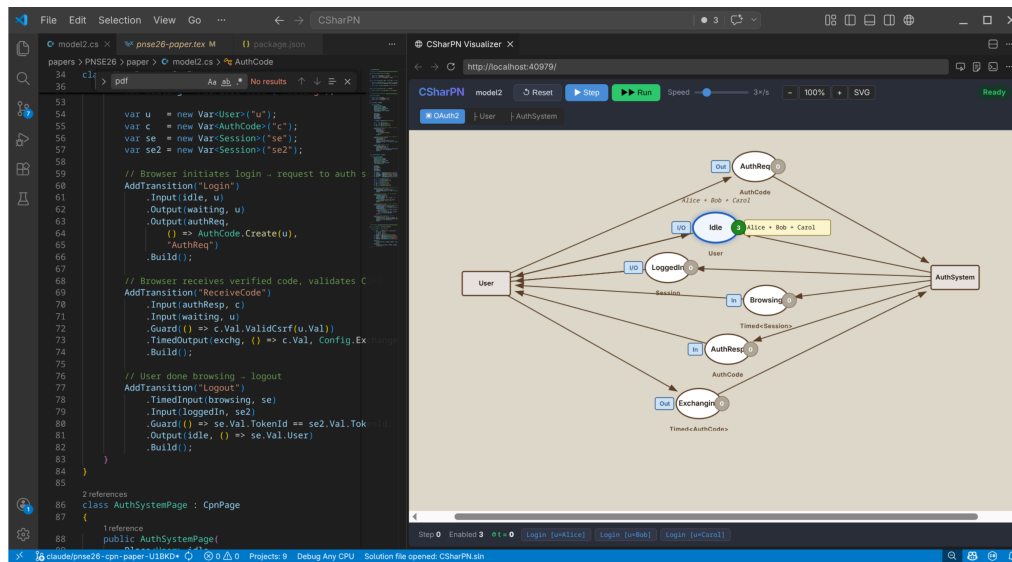
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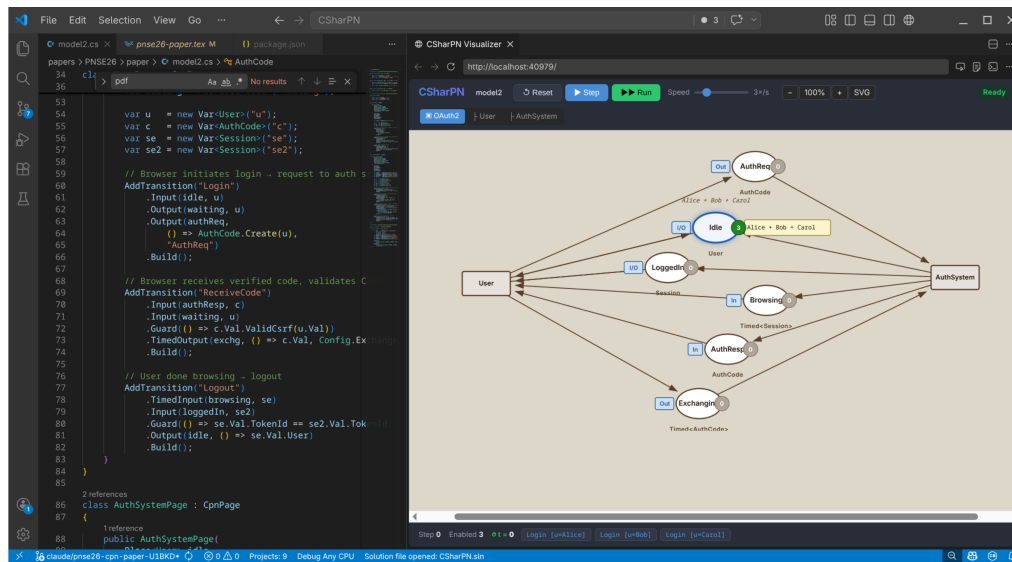
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- **Hot reload**: save .cs → recompile → reload, marking preserved



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key insight

When the inscription language is the host language, the model *is* the program.

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No code-generation step. No generated artefact, no model/implementation sync problem — the gap is zero by construction. (Caveat:

only when the deployment target is .NET.)

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Takeaway: when the inscription language is C#, the full power of a mature programming language — type system, libraries, tooling — is available to the CPN modeller at no extra cost.

Thank you

Questions?



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github.com/simontjell/CSharpN.public

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